NLP-based Sentiment Analysis of Bilingual Product Reviews with Explainable AI for Enhanced Interpretability

ABSTRACT

In this study, we explored Bengali and English e-commerce product review, automated sentiment analysis with traditional deep learning and modern transformer models, and Explainable AI for model interpretability. Review corpora were gathered from online e-commerce websites and preprocessed with language detection, cleaning, tokenization, and normalization; a hand-corrected segment of the corpus (≈15,000 manually corrected Bangla reviews) was employed for experimental testing along with English test data. Two model families were utilized and compared: deep learning baselines (CNN, LSTM, BiLSTM, CNN–LSTM) and fine-tuned transformers (XLM-RoBERTa, BanglaBERT for Bengali; XLM-RoBERTa, RoBERTa-Sentiment for English). Performance was assessed on accuracy, precision, recall, and F1-score; transformer models attained best (BanglaBERT ≈ 91.32% and RoBERTa-Sentiment ≈ 88.96% on critical metrics). LIME was utilized to offer local, human-interpretable explanations of individual predictions, enhancing transparency and trust. The research proves that transformer-based methods work extremely well for multilingual sentiment analysis and also provides empirical knowledge for e-commerce intelligence as well as NLP in low-resource languages.

Keywords - Deep Learning, Transformer, Explainable AI, LIME, CNN, LSTM, BiLSTM, XLM-RoBERTa, BanglaBERT, RoBERTa-Sentiment.

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Chapter 1

Introduction

The importance of sentiment analysis in e-commerce is briefly discussed in this introductory section, which also explains the focus on Bengali and English product reviews. We also describe the study's motivation, objectives, research gap, and models used.

1.1 Introduction

Over the past few years, e-commerce has become a successful business platform. People can buy all kinds of products from e-commerce websites at any time. Online e-commerce prohibits direct product purchases based solely on visual inspection. After making a purchase, everyone gives an opinion as a review, and this review can increase or decrease the sales of the product. An e-commerce website has a lot of products, and these generate a huge amount of reviews. It is very time-consuming or almost impossible for humans to process such a large amount of data. Therefore, we can analyze such a large amount of data with the help of machines. The best way to analyze them is natural language processing (NLP).

Natural language processing is a part of data science, where we train machines in natural human language, and it gives us a human-understandable output. Sentiment is a major segment of NLP, which is usually classified as positive, negative, or neutral. However, with consumers now reaching out across a multitude of languages, sentiment analysis must extend into multilingual and domain-specific text. Bengali, being one of the most popular languages spoken worldwide, would present unique quandaries concerning sentiment analysis: high linguistic complexity, rich morphology, and lack of labeled data. On the other side, English, apart from being a well-resourced language, also proves to be beleaguered because of its evolution in the online-based communication world.

People in Bangladesh like to write reviews very quickly and in short. They write reviews in Bengali, English, a mix of Bengali and English (code-mixed), and Banglish (Bengali sentences written using English alphabets). It is very easy to work with normal Bengali reviews, but Bangla-English (code-mixed) and Banglish are difficult to work with. It is not possible to normalize these types of data with any model or program. We have to convert these into Bengali one by one manually.

In these studies, we applied deep learning and transformer models for our sentiment analysis with both Bengali and English datasets. Two types of models are implemented. First, we implemented deep learning models such as CNN, LSTM, BiLSTM, and hybrid CNN-LSTM models[1]. Second, we utilized fine-tuned

transformer models, including XLM-RoBERTa and BanglaBERT for Bengali, as well as XLM-RoBERTa and RoBERTa-Sentiment for English [2][3]. Thus, we employed surface techniques to evaluate their effectiveness in classifying diverse languages and cultural contexts with differing sentiments. LIME was used to enhance the interpretable quality of our sentiment classification systems[4]. By locally approximating the intricate model by a less complex interpretable substitute model, it provides a post-hoc explanation by outlining the decisive factors affecting the predicted sentiment.

- E-commerce reviews result in huge amounts of multilingual data (Bengali, English, code-mixed, Banglish), making manual analysis quite impossible; NLP and sentiment analysis help determine whether opinions are positive, negative, or neutral.
- Deep learning models (CNN, LSTM, BiLSTM, CNN-LSTM) and transformer models (XLM-RoBERTa, BanglaBERT, RoBERTa-Sentiment) were applied for sentiment classification in Bengali and English datasets.
- LIME was utilized to make the model more interpretable by giving straightforward, human-readable explanations for the predictions. LIME detects and points out the most influential features on the sentiment classification for every review.

After completing every step, we compared the model results and found out the best model. We utilized the Explainable AI (XAI) to gain deeper insight into Bengali and English words.

1.2 Motivation

The growth of e-commerce sites generates huge amounts of user content in the form of product reviews. The reviews provide valuable insights into customer satisfaction and product quality; however, when produced in large volumes, they cannot be analyzed manually. Hence, there arises a big computational incentive for building automated systems that would be able to efficiently analyze large-scale textual data. Sentiment analysis would be the perfect solution to extracting meaning from this data so they can be analyzed for opinions and emotions, hand-in-hand with NLP and deep learning models.

Since English data have been extensively studied in sentiment analysis, research on Bengali text data remains limited, despite Bengali being one of the most widely spoken languages globally. By setting the stage, this work contributes to the advancement of Bengali NLP research as well as strengthens the multilingual sentiment analysis frameworks. Solving this problem adds to the businesses' side in that it contributes to better decision-making while gaining a better understanding of their consumers and, on the flip side, gives us a chance to gain further expertise in NLP, deep learning, and explainable AI.

1.3 Objectives

In this study, the objectives of this research are outlined below to highlight the key steps undertaken to achieve effective multilingual sentiment analysis of e-commerce reviews.

- To collect and preprocess Bengali and English e-commerce product reviews.
- To apply and evaluate deep learning, transformer models.
- To integrate Explainable AI (LIME) for model interpretability.
- To identify the most appropriate model for our study.

1.4 Methodology

The research followed a systematic approach for sentiment analysis of e-commerce product reviews in Bengali and English. To begin with, review data were collected from online sources, followed by steps of preprocessing: text cleaning, tokenization, stop word removal, and encoding. The processed data would then be used for training and testing various models. Deep-learning-based models-CNN, LSTM, BiLSTM, and a hybrid CNN-LSTM-are implemented to capture semantic and contextual patterns. Besides those, transformer-based architectures-XLM-RoBERTa and BanglaBERT for Bengali; and XLM-RoBERTa and RoBERTa Sentiment for English-were employed for enhanced contextual representation. Also from an interpretable standpoint, the authors used Explainable AI techniques with LIME to explain the model prediction. Finally, all models were ranked on the basis of performance metrics to check for the best approach toward a multilingual sentiment analysis.

1.5 Project Outcome

The outcomes of this research are expected to provide both practical benefits and academic contributions. The key outcomes include:

- Development of a bilingual sentiment analysis model for both Bengali and English.
- Comparison between deep learning and transformer-based models to find out the best result.
- Contribution to low-resource languages like Bengali.
- Application of Explainable AI (XAI) to improve model predictions.
- Lastly, understanding customer satisfaction and improving decision-making.

1.6 Organization of the Report

This thesis is divided into six chapters, each chapter is a key component of the research.

Chapter 1—Introduction presents the background of the study, the motivation, the problem statement, the objectives, and an overview of the research. We cover the whole study in this section.

Chapter 2—The background part covers the literature review, similar works, and gap analysis of the study.

Chapter 3—The methodology shows the implementation of the model, system design, requirements, and diagrams.

Chapter 4—Define the environment setup and result analysis. How the model is working, predicting output, and comparing model performances.

Chapter 5—This chapter covers the ethical data used for society and sustainability through e-commerce insights, efficient project management with low costs, and solving complex engineering problems.

Chapter 6—The last chapter of the paper, which is the conclusion. Described limitations and future work of the study.

Chapter 2

Background

In this chapter, we are going to know about the literature review, similar works, related works and the gaps we found in other's work.

2.1 Introduction

The background of this research stems from the upward curve of sentiment analysis, which is a branch of natural language processing that aims to extract opinions and emotions from the text. As people rely more and more on e-commerce, large-scale analysis of customer reviews provides an essential task toward understanding consumer behavior and the performance of products. In this chapter, the theoretical basis of sentiment analysis, NLP methods, deep learning, and transformer models, along with the related work in English and Bengali, has been attempted to be laid down.

2.2 Literature Review

Table 2.1: Similar and Related Literature Reviewed.

Author (s)	Year	Title	Methodology	Key Findings
S. S. Shanto et al.[5]	2023	Binary vs. multi- class sentiment classification for Bangla e-commerce product reviews: A comparative analysis of machine learning models	Machine Learning models	Binary classification perform better in machine learning models
R. Chowdhury et al. [6]	2022	Rate insight: A comparative study on different machine learning and deep learning approaches for product review	Machine Learning and Deep Learning	LSTM-based approaches shows better result then ML models

		rating prediction in Bengali language		
O. Bellar et al. [7]	2024	Sentiment analysis: Predicting product reviews for E-commerce recommendations using deep learning and transformers	Deep Learning and Transformer	Transformer-based deep learning models enhance the accuracy of sentiment analysis
T. Alsaedi, M. et al. [8]	2024	Sentiment mining in E-commerce: The transformer-based deep learning mode	Transformer based Deep Learning	Transformer-based deep learning models performs better then traditional methods
N. R. Bhowmik et al. [9]	2021	Sentiment analysis on Bangla text using extended lexicon dictionary and deep learning algorithms	Lexicon and Deep Learning	Combination of Bangla lexicon dictionary and deep learning improve accuracy.
M. S. Islam et al. [10]	2023	Sentiment analysis on Bangla food reviews using machine learning and explainable NLP	Machine Learning and Explainable AI	Explainable AI enhance machine learning model accuracy
S. N. Nobel et al. [11]	2024	Hybrid CNN LSTM approach for sentiment analysis of Bengali language comment on Facebook	Hybrid CNN LSTM	Hybrid model effective for sentiment classification accuracy
P. Savci et al. [12]	2023	Prediction of the customers' interests using sentiment analysis in e-commerce data for comparison of Arabic and English	Machine Learning algorithms	Multilingual e-commerce data can play a significant role in model implementation

2.2.1 Similar Applications

M. A. Rahman et al. [13] analyzed sentiments of Bangla and Romanized Bangla text using transformers. They created three datasets for Bangla, Romanized Bangla and combined with a total of 6268 data. XLM-RoBERTa performed best for combined data, while ANN performed best for Romanized data. M. N. Hoque et al. [14] did sentiment analysis on under-resource Bengali language. With a dataset of 11,807 comments, they applied five transformer models and they are mBERT, BanglaBERT, Bangla-BERT-Base, DistilmBERT, XLM-RoBERTa. While their proposed model shows better performance with 95.97% accuracy and 95.96% F1-score. After that, M. Hoq et al. [15] also did sentiment analysis but using Deep Learning models and focused on emotion. They applied four hybrid CNN-LSTM models with different word embeddings (Word2Vec, GloVe etc.). The best model was CNN-LSTM with Word2Vec with an accuracy of 90.49% and 92.83% F1-score.

2.2.2 Related Research

Recent studies in sentiment analysis have greatly enhanced understanding of user opinion in the Bangla language in a range of applications, from e-commerce to social media. Studies have investigated deep learning methodologies in Bangla sentiment classification with positive findings [15][16]. The use and availability of big datasets, as made by [17] and [18], allowed more effective model and evaluation training, and approaches based on machine learning have also been effective in classifying Bangla reviews [19]. Lexicon-based tools, as shown by BengSentiLex, have also facilitated improved results in low-resource environments [20]. Transformer-based architectures, including recent unsupervised and aspect-adaptive ones, have also enhanced accuracy and compensated for problems associated with data imbalance aspects [22]. Furthermore, experiments on e-commerce domains suggest the potential value of sentiment analysis in customer satisfaction and multilingual user understanding fields [23][24]. Large datasets created with particular niche problems, as with cricket sentiment on Bangla social media, have also expanded research application possibilities [25]. These all register as indicative of the field-wide implications and potential growing impact of sentiment analysis in the Bangla language processing field.

2.3 Gap Analysis

In this part of the study, we are going to analyze, observe and evaluate existing works. There are a lot of papers about sentiment analysis. It will help us to complete our work.

Table 2.2: Gap Analysis of reviewed papers.

Paper	Approaches Used	Dataset	Limitation/Gaps
Taneja et al. [22]	Transformer (Unsupervised)	23,486 e-commerce reviews	Only English
Ahmed et al. [2]	BERT-based (Transformer)	1000 Bengali reviews	No comparison with other models, small dataset
Shan et al.[23]	Transformer-based bilingual model	Multilingual e-commerce reviews	No specific analysis in Bengali language
Alsaedi et al. [8]	Transformer-based deep learning model	2270 customers e-commerce reviews	Lack of insights for multilingual dataset, smaller dataset
Nobel et al. [11]	Deep Learning (CNN + LSTM)	50,000 Bengali Facebook comments	No Transformer model used

In the end we can see, no one used Explainable AI(XAI) in their implementation.

2.4 Summary

In summary, while deep learning and transformer-based approaches have substantially advanced sentiment analysis, there are important gaps with regard to Bangla and low-resource languages in general. Most existing study mainly focuses on high-resource languages or does not offer comprehensive support for multilingual and Romanized Bangla. A discussion of gaps, in turn, highlights a need for a larger array of datasets, efficient mechanisms for benchmarking, and domain-specific models capable of addressing the specific linguistic challenges involved in sentiment analysis in Bangla. Such conclusions lay foundations for later research directions aimed at advancing the accuracy and relevance of sentiment detection in less represented languages.

Chapter 3

Research Methodology

This chapter defines the methods applied to conduct the research and implement the designed system. It defines the dataset, data preprocessing techniques, model selection, requirement analysis, and design requirements, accompanied by supportive diagrams and project planning. These subsections in aggregate provide a clear-cut overview of how the research was structured and how the system was planned.

3.1 Methodology/Requirement Analysis & Design Specification

3.1.1 Overview

This project is proposed to create and implement a multilingual (Bangla, English, Banglish) sentiment analysis platform for e-commerce product reviews through the union of research methodology and software development. At the research level, the effort explores both transformer and deep learning-based architectures, modifying them for application on Bangla, English, and Banglish reviews collected from Daraz the largest e-commerce website in Bangladesh. At the project level, the system was carefully developed through requirement analysis, task allocation, and multiple-layered design specifications in order to remain reliable, scalable, and easy to use.
The general approach is the merger of two elements: one, a methodology oriented towards research that deals with dataset preparation, preprocessing, model selection, training, and testing; and two, system-oriented design that dictates functional and non-functional specifications, data flow, task allocation, and user interface issues. Together, they guarantee that the project not only possesses an academic impact through the demonstration of comparative model performance but also presents a working web-based system with the capacity for real-time sentiment classification.

3.1.2 Proposed Methodology/ System Design

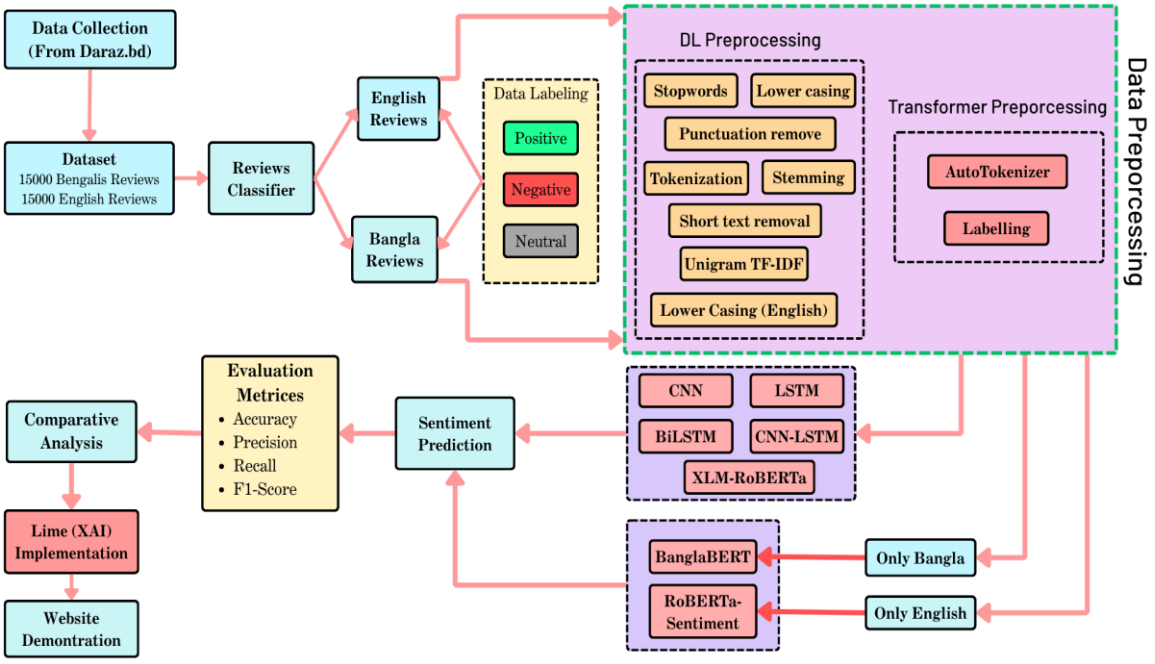

Figure 3.1: The proposed methodology

This proposed research methodology develops a bilingual sentiment analysis approach for e-commerce reviews collected from Daraz (Bangladesh) in Bangla, English, and Banglish. Having taken on a manual sentiment labelling approach, two preprocessing techniques were implemented: transformer-specific preprocessing through AutoTokenizer and deep learning preprocessing with tokenization, stemming (Bangla stemmer for Bengali reviews and NLTK for english reviews), punctuation removal, emoji removal, and unigram TF-IDF. A number of deep learning and transformer-based models were compared and contrasted in both English and Bengali reviews, namely CNN, LSTM, BiLSTM, CNN-LSTM (Hybrid) and XLM-RoBERTa. But for English-specific evaluation RoBERTa-Sentiment, while BanglaBERT was utilized for Bangla-specific reviews and Banglish texts (after transliteration) for production deployment. Accuracy, precision, recall, and F1-score were analyzed for the evaluation. Finally, explainability was achieved by applying LIME, and the integrated system was made available through a web-based interface

3.1.3 Functional and Nonfunctional Requirements

Considering both functional and non-functional requirements assures the system's effectiveness and productivity. Functional requirements identify the operations the system ought to perform, for example, sentiment analysis, language review preprocessing, and a web page display of results. Non-functional requirements influence the performance of the system, for example, how fast it should be, its scalability, usability, and reliability. Altogether, they contribute to the strengthening of my work structure, efficacy, and preparedness for application in the real world. Cumulatively, these requirements strengthen the research project by:

- Providing structure and clarity by separating between the "must-have features" and the "quality features" so that nothing is overlooked.
- Improving efficiency: the models and APIs were created to achieve response-time and accuracy requirements.
- Making it usable: the system is an operational online application that can be utilized for practical applications, not only an academic prototype.

Functional Requirements:
The functional requirements specify the core features and behaviors of the proposed sentiment analysis system. These are the operations that the system must perform to meet the project objectives.

- Input of Reviews by User: The system must have a feature that will allow the user to enter product reviews in Bangla, English or Banglish through the web user interface. Users should be able to paste or type multiple comments at a time.

- Detection of Language and Preprocessing: The system must automatically detect the input language (is it in bangla, english or banglish). Then it must normalize Banglish text into Bangla script and the preprocessing should include stopword removal, stemming, punctuation removal, tokenization, lowercasing (for English reviews only), and short text removal (but careful about reviews sentiment).

- Sentiment Analysis: The working system must be able to analyze each review into one of three sentiment labels: Positive, Negative and Neutral. Deep learning models (CNN, LSTM, BiLSTM, CNN-LSTM hybrid) and fine-tuned transformer models (BanglaBERT, RoBERTa, XLM-RoBERTa) must be supported. The system must illustrate a confidence score with each prediction alongside a detailed score section.

- Model Selection and Prediction: The backend part must allow conditioning between models for implementation (e.g., BanglaBERT for Bangla reviews, RoBERTa for English). The /predict API endpoint must accept a review and return the sentiment results in JSON (JavaScript Object Notation) format.

- Visualization of Results: The frontend part must display the predicted sentiment and its confidence level of trust. The UI should be able to illustrate

a simple interface showing whether a review is positive, negative, or neutral.

- Dataset Handling: The research work must support training and testing on the collected dataset of 15,000+ reviews. The system must split data into training and testing sets (default 80/20).

- Deployment and Web Integration: The FastAPI backend must provide endpoints accessible via HTTP. The React.js frontend must send POST requests using Axios and render results dynamically.

Non-functional Requirements:

Non-functional requirements in software engineering refer to the characteristics of a software system that are not related to specific functionality or behavior. They describe how the system should perform, rather than what it should do. This article focuses on discussing non-functional requirements in detail. [3.1]

- Performance: The sentiment predictions from analysing the reviews must be returned within 2 to 3 seconds for every single review under normal data load. The system must support at least 50 concurrent requests without interruption or failure.

- Scalability: The architecture of the research work must allow additional models to be integrated without major changes to the frontend. The backend should support the future extension to larger datasets, for example we have a total of 75000+ reviews but we used only 15000 reviews in this research work. So in the near future the dataset will be increased to 75000 reviews. So the backend or the system should be able to handle these future trends.

- Reliability: The system must maintain consistency in the performance and availability during usage. The backend must be able to handle invalid or inconsistent inputs gracefully (e.g., empty text, unsupported characters).

- Usability: The user interface must be simple and interactive and can be aesthetic in design, allowing non-technical users to test reviews easily. The UI should have clear instructions. Each section should be visually appealing for example red color for Negative, Green For Positive etc.

- Maintainability: The system must use simple code structure to allow updates of preprocessing steps, models, or frontend components independently. Documentation of APIs and model usage should be provided.

- Security: The API must enforce Cross-Origin Resource Sharing (CORS) rules to allow safe communication between frontend and backend.

- Portability: The system must be deployable on standard operating systems (Windows, Linux). The backend must support containerization or cloud deployment in the future.

3.1.4 Context Diagram

Context Diagrams are high-level visual representations that show the interactions between a system being developed and its external entities, such as users, other systems, or processes. They provide a big-picture view of how the system fits into its environment without diving into the internal details of the system itself. [3.2]

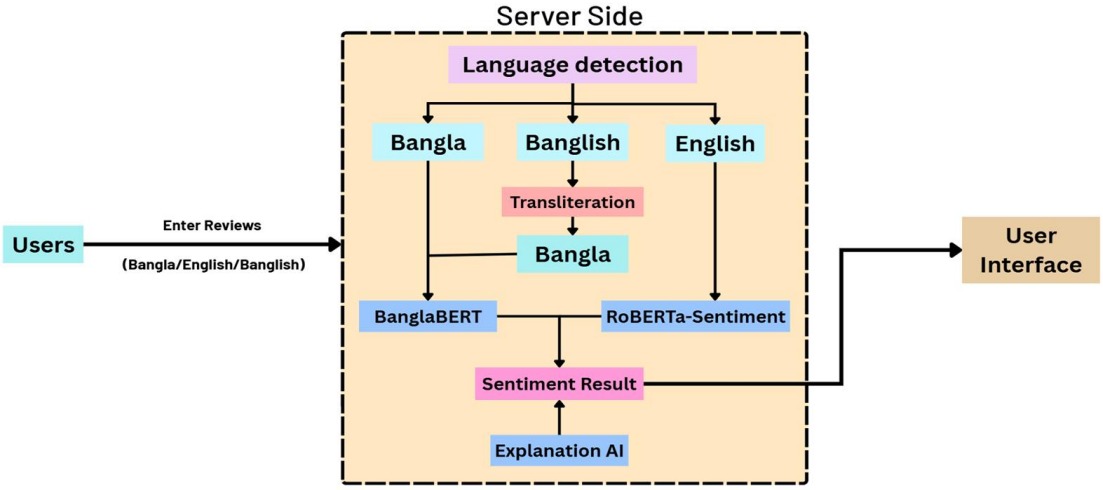

Figure 3.2: Context Diagram

The context diagram shows how the system users enter Bangla, English, or Banglish reviews through the web user interface. The system processes the inputs, applies deep learning or transformer models, and returns sentiment predictions with confidence scores, using datasets and models for training and testing.

3.1.5 Data Flow Diagram

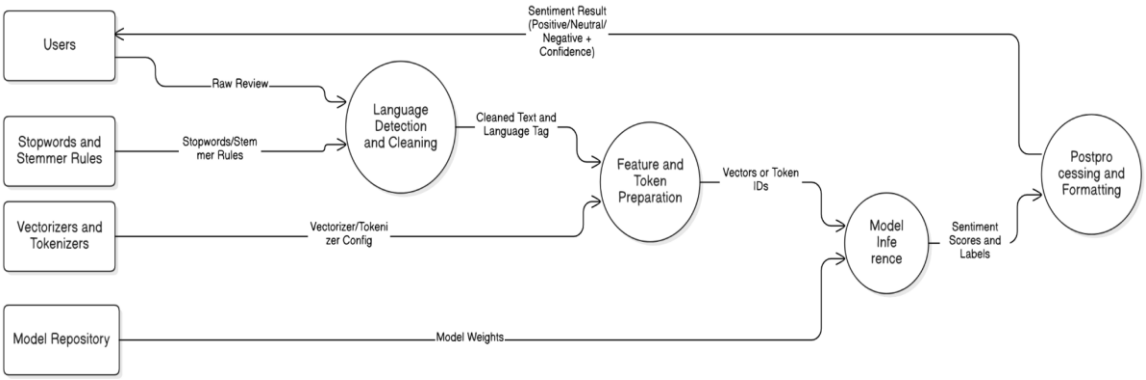

Figure 3.3: The proposed Data Flow Diagram

This Data Flow Diagram precisely shows how the sentiment analysis system works. The user first enters review and it gets detected with language detection and Cleaning, filtering out stopwords and Banglish text normalisation. After cleaning, the input is propagated to Token Preparation (Autotokenizer) for transformer tokenization or TF-IDF. Then, in order to produce predictions, Model Inference uses fine-tuned CNN, LSTM, BiLSTM, or transformer models. The output is then

organized into user-oriented sentiment tags with confidence scores through post-processing.

3.1.6 UI Design

The proposed User Interface and Initial design the sentiment analysis system was kept straightforward, interactive, and intuitive to ensure ease of use for both technical and non-technical users. The interface gives a contemporary and responsive feel, having been developed with Vite+React.js for rendering speed and TailwindCSS and DaisyUI for styling. With just one click, the users are able to analyze reviews of products in Bangla, English, or Banglish and get the sentiment label and confidence score in percentage. It is suitable for real-time testing and hands-on demonstration during thesis defence due to its observable visual indicators and user-friendly design elements.

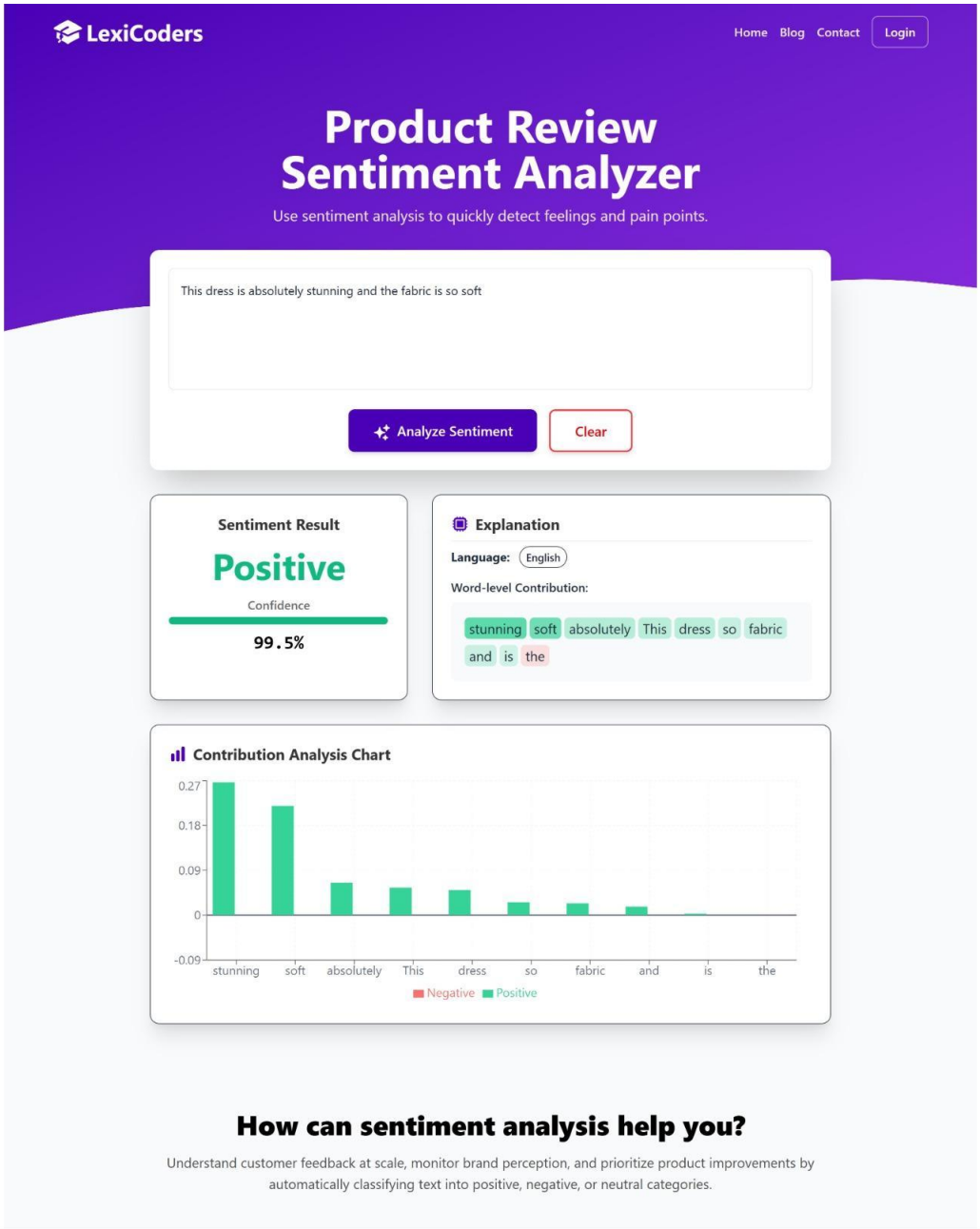

Figure 3.4: The proposed and Initial User Interface

3.2 Detailed Methodology and Design

The proposed research system is made to perform sentiment analysis of Bangla/English product reviews collected from online e-commerce sites. The research work uses a hybrid approach between conventional deep learning methods (LSTM, CNN, BiLSTM, CNN-LSTM) and transformer-based (BanglaBERT, RoBERTa) systems in order to deliver guaranteed performance on diverse linguistic inputs. The process involved careful consideration of various alternatives prior to the ultimate choice of the adopted methodology.

3.2.1 Alternate Solutions Considered

- Classical Machine Learning Models (SVM, Naïve Bayes, Logistic Regression): They were initially considered because they are easy and fast to train. They are good at handling TF-IDF features. However, they fail to maintain semantic meaning and context relationships that are important in sentiment analysis, especially for morphologically rich languages like Bangla.

- Rule-based Sentiment Lexicons: A second alternative approach was to build a Bangla–English sentiment lexicon and perform polarity scoring. While this is a lightweight and explainable approach, it breaks down with mixed-language reviews (Banglish) and does not capture contextual cues like sarcasm or negation.

3.2.2 Selected Solution and Justification

After evaluating alternatives, a hybrid deep learning and transformer-based design was selected:

- For deep learning, the following models were used on TF-IDF features as baselines: CNN, LSTM, BiLSTM, and CNN-LSTM. These capture sequential and local patterns, providing performance levels against which advanced models are compared.

- For transformers, BanglaBERT (A BERT based model pre-trained on large datasets of Bengali text) was applied to Bangla product reviews, RoBERTa to English reviews, and XLM-RoBERTa for multilingual input. Transformers were used because of their ability to model long-distance dependencies, contextual word representations, and improved performance on multilingual NLP tasks.

- A modular preprocessing pipeline was employed to handle stopword removal, stemming, normalization, and tokenization so as to remain compatible with both DL and transformer pipelines.

3.2.3 Dataset

The dataset that is used in this research work was collected from Daraz (an e-

commerce website in Bangladesh probably the largest e-commerce platform) through web scraping technologies and manual scraping. This dataset consists of about 15000 Bengali product reviews. The dataset was almost balanced to include an equal distribution of three sentiment labels : Positive (5,147 reviews), Negative (5066 reviews), and Neutral (4,821 reviews). The original dataset contains Bangla, English and Banglish Reviews. But the final dataset contains only 15000 pure bangla reviews which were manually processed and labelled. The dataset was splitted into train and test sets using 80/20 rule. The training set was used for model fine-tuning, while the test set was used to evaluate performance.

Table 3.1: Sample of Dataset

Review	Sentiment
অসাধারণ অস্থির	Positive
অস্থির একটি প্রোডাক্ট। চোখ বন্ধ করে নিতে পারেন।	Positive
Keeps warm and uncomfortable at the same time.	Neutral
Action should be taken against them!!	Negative

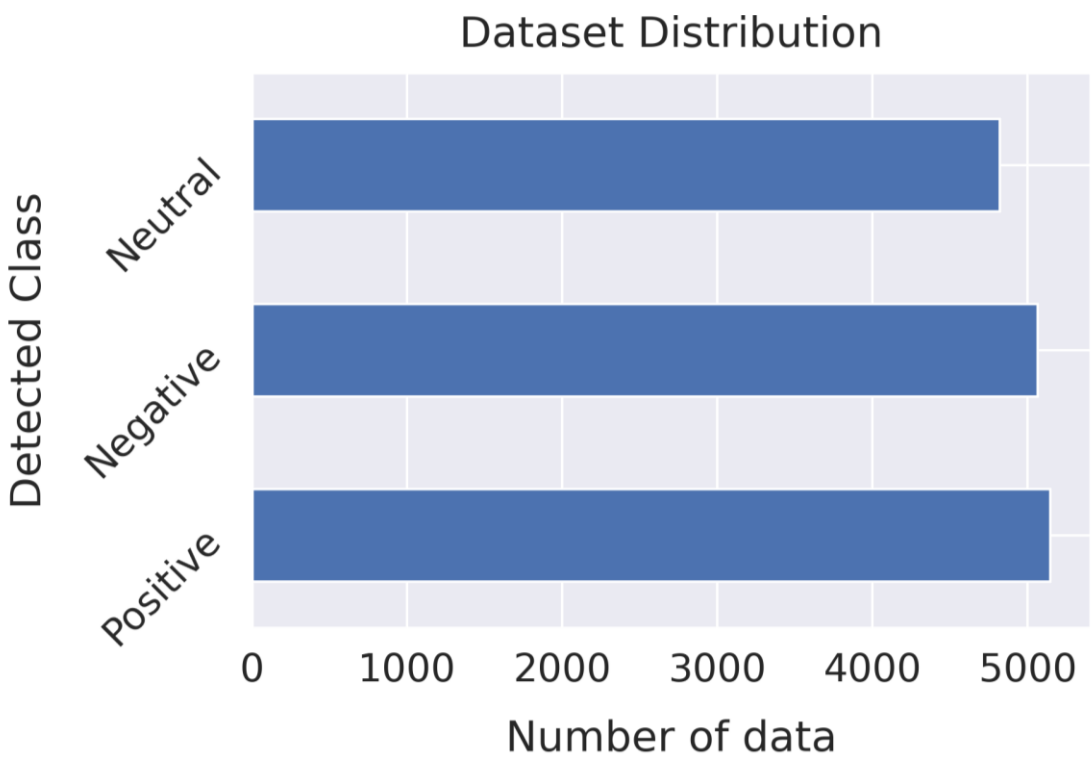

Figure 3.5: Dataset Distribution Chart

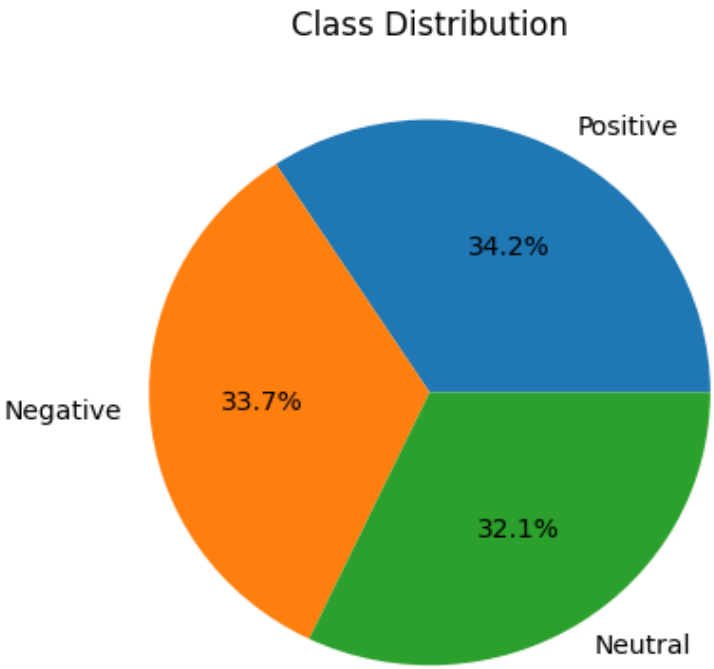

Figure 3.6: Class Distribution Pie Chart

3.2.4 Data Preprocessing

Preprocessing was required to clean noisy text and prepare it for transformer models and deep learning. Two distinct pipelines were there:

Preprocessing for Deep Learning Models:

Stopword Removal: Bangla and English common stopwords were removed using a Bangla stopwords list and NLTK stopwords for English.

Table 3.2: After cleaning data.

Original	Cleaned
অনেক ভালো পন্যটি ধন্যবাদ দারাজকে।	ভালো পন্যটি ধন্যবাদ দারাজকে

Stemming: Bangla text was stemmed with Bangla Stemmer. English text was stemmed with the NLTK stemmer.

Removal of Punctuation: All punctuation characters were removed.

Lowercasing: English text was downcased to lowercase for normalization, and Bangla text was left in original form.

Short Text Removal: Very short reviews (below three words) were removed to reduce noise. But be careful about the meaning of the text. Like the word “NO” is very short

but contains significant sentiment.

Vectorization: Preprocessed text was mapped into TF-IDF vectors (unigrams), capturing word salience in the corpus.

Table 3.3: N-grams for the reviews based on TF-IDF

Review	Unigram	Bigram	Trigram
ঠিকঠাক ছিল সেলার ভাই আচরণ ভালো ছিল ফাস্ট চার্জ চার্জার কিছুটা গরম হয়	('কঠ',1)	('কঠ আচরণ',1)	('কঠ আচরণ গরম',1)
	('আচরণ',1)	('আচরণ গরম',1)	('আচরণ গরম হয',1)
	('গরম',1)	('গরম হয',1)	

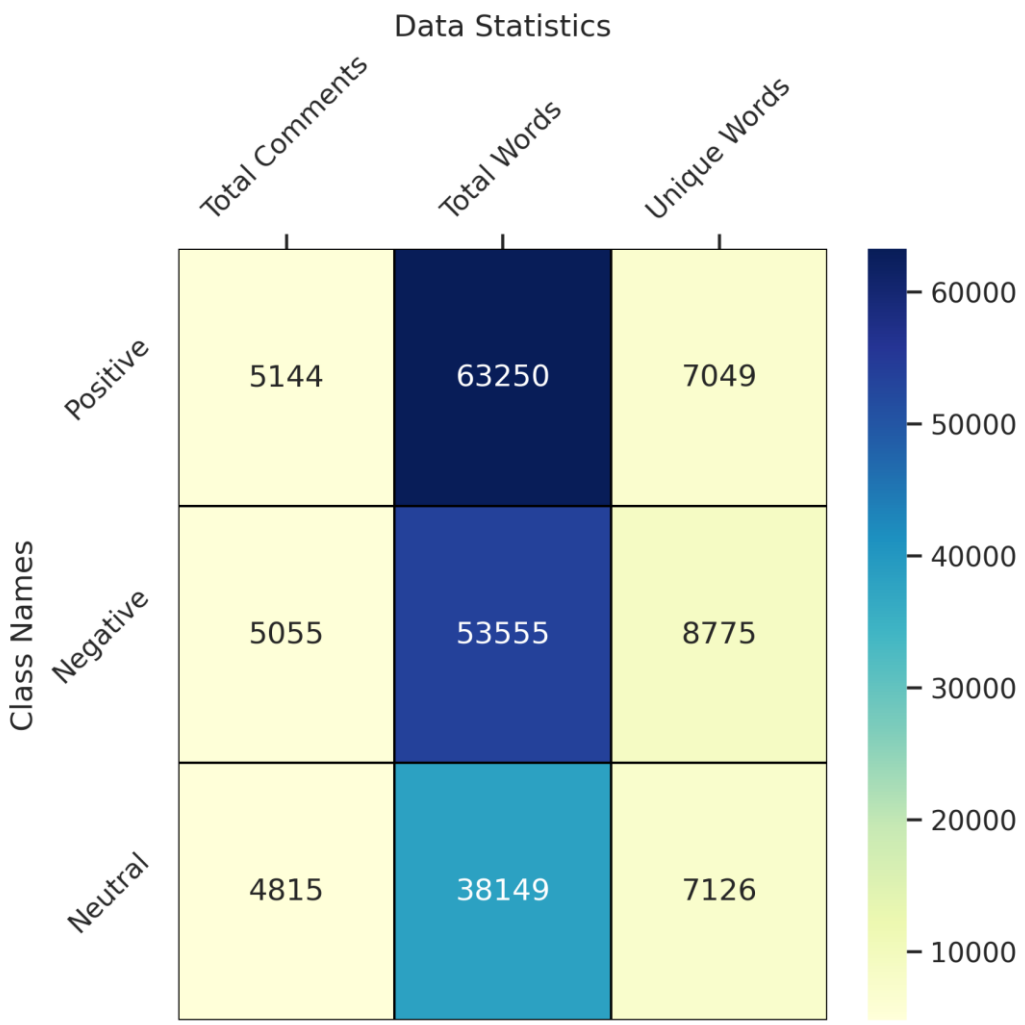

Figure 3.7: Data Statistics Bangla

Preprocessing for Transformer Models:

Tokenization: Hugging Face AutoTokenizer was used for BanglaBERT, RoBERTa, and XLM-RoBERTa.

Label Encoding: Sentiment classes (Positive, Negative, Neutral) were encoded into integers.

Padding and Truncation: Tokenized sequences were standardized to fixed length for batch processing.

3.2.5 Models Description

In this work, two model categories—deep learning models and transformer language models—were implemented and tuned to execute effective sentiment categorisation. The two categories were selected because they could be compared between modern attention-based models and conventional sequence models, and each family has a distinct strength in processing text data.

Convolutional Neural Network (CNN)[26]: CNNs are widely used in NLP tasks like sentiment analysis, text classification Though originally designed for computer vision, CNNs are also widely used for natural language processing. Simply put, just think of any text sequence as a one-dimensional image. In this way, one-dimensional CNNs can process local features such as n-grams in text. [3.3]

Long Short-Term Memory (LSTM)[27]: LSTM (Long Short-Term Memory) is an advanced version of RNN designed to remember information for long periods. Unlike traditional RNNs, LSTMs can retain context over longer sequences, making them ideal for text-related tasks. Unlike traditional models, LSTM understands the order of words, which is crucial in text like "not good" vs. "good." [3.4]

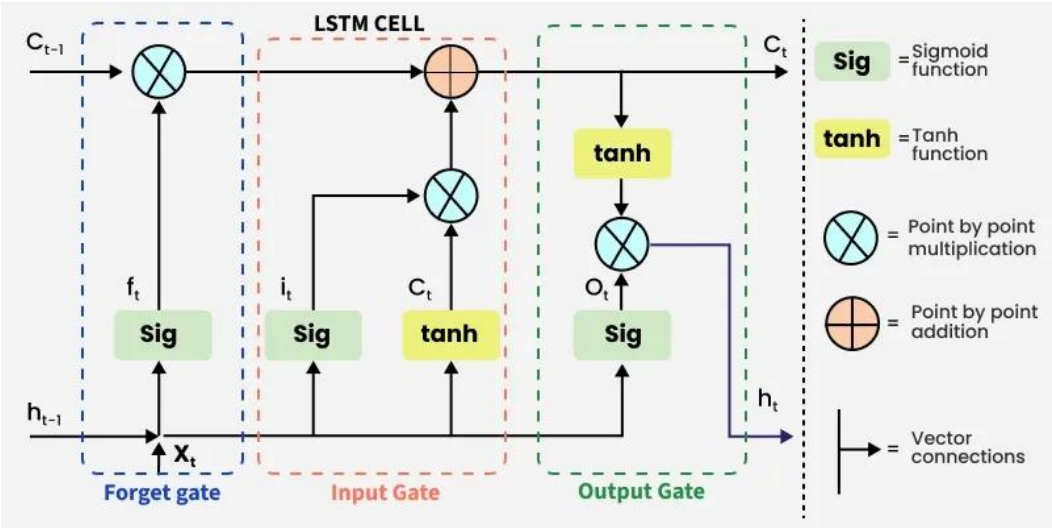

Figure 3.8: LSTM Architecture

Bidirectional Long Short-Term Memory network(BiLSTM)[28]: BiLSTM, or Bidirectional Long Short-Term Memory, is a refined type of recurrent neural networks (RNNs) that processes input sequences in two directions, namely, forward and backward. Unlike ordinary LSTM, whose history is only summarized in the previous history, BiLSTM encapsulates both previous and succeeding history and, therefore, enhances its performance in application domains such as sentiment analysis, text classification, and language modeling. Its capability of capturing all the context in a sentence considerably enhances its efficiency in processing sequence text data.

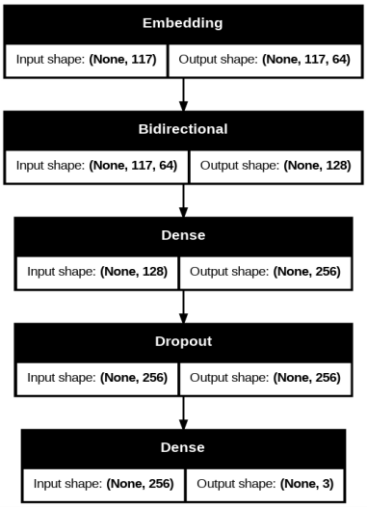

Figure: 3.9: BiLSTM Architecture

CNN-LSTM[29}: Hybrid CNN-LSTM model refers to a deep network structure that integrates Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks to achieve improved performance in text-based operations like sentiment analysis. CNN layers are efficient in extracting local features and patterns from text (e.g., combinations of words or key phrases), whereas LSTM layers learn long-term dependencies and context relationships between sequences. By combining the two architectures, the hybrid model is capable of utilizing CNN's feature extraction capability as well as LSTM's capability in extracting the global context and thereby achieving better and stronger sentiment or emotion classification.

Transformer Models: Transformers utilize the self-attention mechanism, which enables each word to consider the importance of every other word in the list. For a given token, the model computes three vectors: query (Q), key (K), and value (V). The attention score is computed as:

$$\boldsymbol{Attention(Q,K,V) = softmax(\frac{QK^T}{\sqrt{dk}})V}$$

Three transformer models were used in this research:

XLM-RoBERTa[30]: XLM-RoBERTa is a multilingual RoBERTa model pre-trained on 100 languages from large-scale CommonCrawl data. It is an extension of RoBERTa and does well in cross-lingual tasks without the need for language-specific tokens, thus being usable for sentiment analysis in low-resource languages such as Bangla.

BanglaBERT[31]: BanglaBERT is a transformer model specially trained on enormous-scale Bangla text. It is meant to improve the linguistic structure understanding of the Bangla language and is superior to multilingual models such as mBERT and XLM-RoBERTa for Bangla-centric NLP tasks like sentiment analysis, text classification, and named entity recognition.

RoBERTa[32]: RoBERTa-Sentiment is an adjustment of the RoBERTa model that has been fine-tuned for use in sentiment analysis tasks exclusively. It utilizes the strong language comprehension ability of RoBERTa and adjusts it to label text as possessing

sentiment labels (positive, negative, or neutral) with great precision, particularly within domain-specific applications such as e-commerce or social media studies.

3.3 Project Plan

The project was implemented in a series of structured phases to allow sequential development, experimentation, and deployment of the multilingual sentiment analysis system. The strategy adopted a research-driven workflow, balancing dataset preparation, model experimentation, and web-based interface implementation.

Phase 1: Literature Review and Problem Analysis

Phase 2: Dataset Collection and Preparation

Phase 3: Data Preprocessing

Phase 4: Model Development and Training

Phase 5: Comparative Analysis and Evaluation

Phase 6: System Implementation (Backend and Frontend)

Phase 7: Documentation and Thesis Preparation

Table 3.4: Gantt Chart of Project Timeline

Phases	4	10	14	16	18	20	22	24
Literature Review and Problem Analysis								
Dataset Collection and Preparation								
Data Preprocessing								
Model Development and Training								
Comparative Analysis and Evaluation								
System Implementation (Backend and Frontend)								
Documentation								

and Thesis Preparation								

3.4 Task Allocation

The project was designed as a collection of well-defined tasks that each had some contribution toward the final development of the multilingual sentiment analysis system. First, effort was given to the problem analysis and literature review, in which research works on sentiment analysis, particularly in Bangla and multilingual, were critically evaluated. This phase helped to identify research gaps and to sharpen the thesis goals.

Then, the focus shifted to data collection and preprocessing. 15,000 reviews of e-commerce were gathered from Daraz in Bangla, English, and Banglish language. Duplicates and nonsense entries were filtered out of the dataset after which class distribution was maintained for a balanced presentation of positive, negative, and neutral reviews.

The second task was data preprocessing, which was necessary for text data to be converted into machine-readable data. Deep learning and transformer models had independent preprocessing pipelines. For deep learning, preprocessing techniques such as stopword removal, stemming, filtering of punctuation, lowercasing, and TF-IDF vectorization were performed. For transformer models, tokenization with Hugging Face AutoTokenizer, label encoding, and padding of sequences were performed.

Post-preprocessing, model training and development were emphasized. Four deep learning models, i.e., CNN, LSTM, BiLSTM, and CNN-LSTM, were trained to establish baseline performance. In parallel, transformer-based models like BanglaBERT, RoBERTa, and XLM-RoBERTa were fine-tuned to achieve more semantic-rich representations. Optimizing accuracy, precision, recall, and F1-score during training.

The output of the models was then compared when being analyzed and judged. Metrics and confusion matrices were established to identify differences in architectures. The comparative analysis demonstrated the advantages of transformer models compared to traditional deep learning models, especially when handling multilingual and mixed-language inputs.

The following was system implementation, where the trained models were deployed within a FastAPI backend for prediction services. A React.js frontend was developed to provide an intuitive interface, themed with TailwindCSS and DaisyUI. The two layers were coupled via REST APIs, where users could input reviews and receive real-time sentiment predictions.

Finally, sufficient time was dedicated to documentations and thesis writings. This activity involved encapsulating the entire process, from dataset creation to

deployment, into a formal report. Additional materials such as context diagrams, DFDs, Gantt charts, evaluation tables, and UI captures were prepared to support the presentation and clarity of the thesis.

3.5 Summary

This chapter provided a general description of the way in which the research and project were structured, from concept generation to system design. The Overview provided the two-goal strategy: an investigation-driven strategy for testing models and a project-driven strategy for building a working application. Functional and non-functional specifications were created to ensure that it was understood what the system must do and how it ought to behave with respect to ease of use, efficiency, and scalability.

To depict interactions, a Context Diagram and Level 0 and Level 1 Data Flow Diagrams were provided, showing the relationship of user input, preprocessing, models, and output. A UI design description identified the availability and ease of the web interface and was supported with screenshots.

The Detailed Methodology and Design part identified dataset preparation, preprocessing strategies, and the comparative model architecture. Deep learning models (CNN, LSTM, BiLSTM, CNN-LSTM) and transformer models (BanglaBERT, RoBERTa, XLM-RoBERTa) were sketched, wherein equations illustrated how they operate with text data. The layered System Design was described with preprocessing, model, API, and presentation layers so that it would be modular and extensible.

Project management aspects were discussed as well. Elements of Project management were addressed as well. A Project Plan with a Gantt chart mapped the timeline from January to August 2025. Task Allocation was described and graphed out, defining responsibility by phase such as data gathering, preprocessing, training, testing, implementation, and documentation.

Together, this chapter combined research rigour with practicable implementation, showing how theoretical methods of sentiment analysis were translated into an actual system. Bringing together diagrams, equations, and planning tools emphasized technical complexity as well as project viability, enhancing overall thesis contribution.

Chapter 4

Implementation and Results

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4.1 Environment Setup

Table 4.1: Environment setup table.

Component	Description
Laboratory	Colab, Jupyter Notebook
Programming Language	Python
Libraries & Frame-works	TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformers, NLTK, SpaCy
Preprocessing Tools	Tokenization, stop-word removal, text cleaning, encoding
Hardware Specification	Intel Core i3 Processor, 8 GB RAM, NVIDIA GPU/Intel inside gpu
Operating System	Windows 10

4.2 Testing and Evaluation/Performance/ Comparative Analysis

4.2.1 Model Evaluation:

The results and discussion are a significant aspect of the research, providing a clear picture of the research outcome. This section of the paper provides a close discussion in light of the data set, the outcome of various models, and the assessment of their corresponding scores.

Accuracy: The accuracy of a model refers to the proportion of correct predictions it

makes, calculated as the ratio of true positives and true negatives to the total number of positive and negative observations.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

Precision: It is made up of the ratio of the number of correct model predictions, which focuses on the precision in relation to positive predictions.

$$Precision = \frac{TP}{(TP + FP)}$$

Recall: Recall indicates how effectively a model can identify all instances of a given class correctly. It is typically calculated as the ratio of true positives to the sum of true positives and false negatives.

$$Recall = \frac{TP}{(TP + FN)}$$

F1-score: This is one of the measures statisticians use when analyzing data. When precision and recall are combined, it forms a hybrid measure known as the F1-score, which is calculated as the harmonic average of precision and recall.

$$F1 = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)}$$

4.2.2 Comparison Model Table:

In table 4.1, we can see a comparison between implemented models. Here we applied Deep Learning and Transformer models. As we can see, CNN performed best with an accuracy of 83.42% in Bengali and 79.72% in the English dataset. After that, from the transformer model, we can see BanglaBERT and RoBERTa-Sentiment outperformed with an accuracy of 91.32% in Bengali and 88.96% in the English dataset.

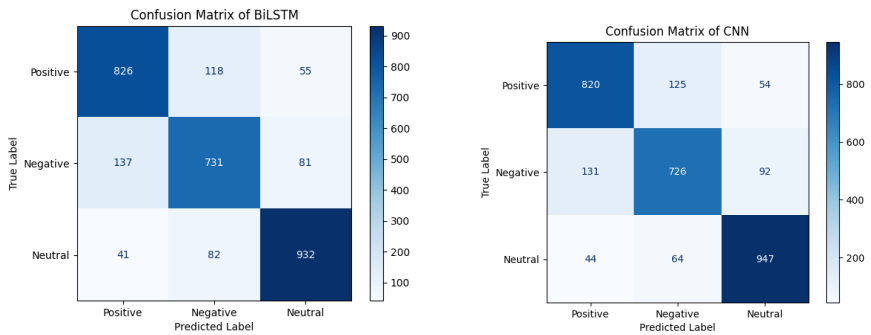
Table 4.2: Comparison table.

	Bengali				English			
Models	**Accuracy**	**Precision**	**Recall**	**F1-score**	**Accuracy**	**Precision**	**Recall**	**F1-score**
CNN	83.42	83.23	83.22	83.22	79.72	80.06	79.37	79.16
LSTM	83.01	82.80	82.78	82.77	78.50	78.52	78.50	78.50
BiLSTM	82.88	82.68	82.68	82.63	78.93	78.93	78.87	78.89
CNN-LSTM	76.02	75.92	75.95	75.93	75.75	75.75	75.78	75.77
XLM-RoBERTa	90.49	90.44	90.40	90.42	88.69	88.72	88.61	88.61
BanglaBERT	**91.32**	**91.30**	**91.32**	**91.29**				
RoBERTa-Sentiment					**88.96**	**88.91**	**88.96**	**88.90**

4.1 Results and Discussion

4.3.1 Confusion Matrix comparison:

A confusion matrix is an effective method to evaluate classifier models. From the confusion matrix, a clear vision of the model's outcome can be achieved, helping determine whether the acquired accuracy is valid. It also allows for identifying and addressing issues such as underfitting or overfitting.



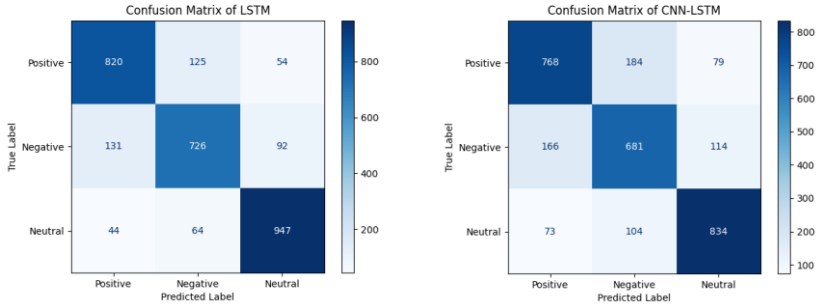


Figure 4.1: Confusion matrix of Deep Learning models in Bengali dataset

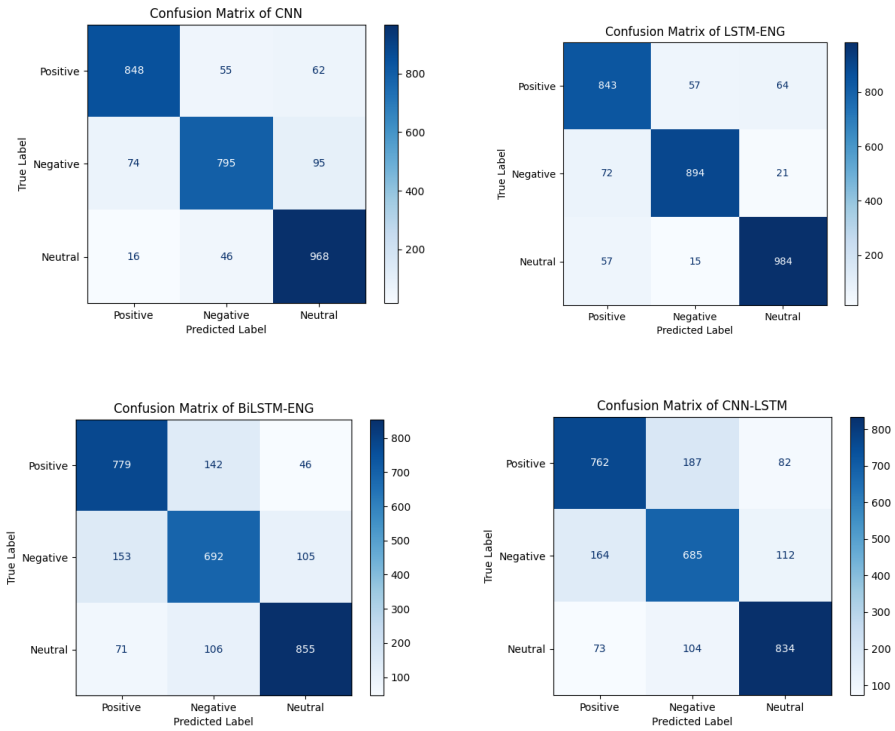


Figure 4.2: Confusion matrix of Deep Learning models in English dataset

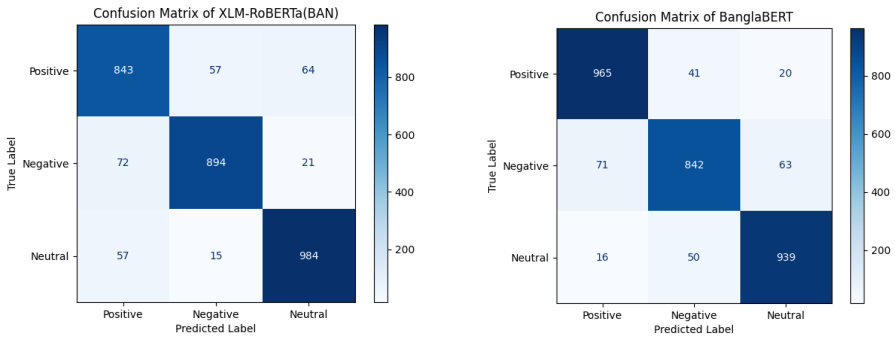


Figure 4.3: Confusion matrix of Transformer-based models in Bengali dataset

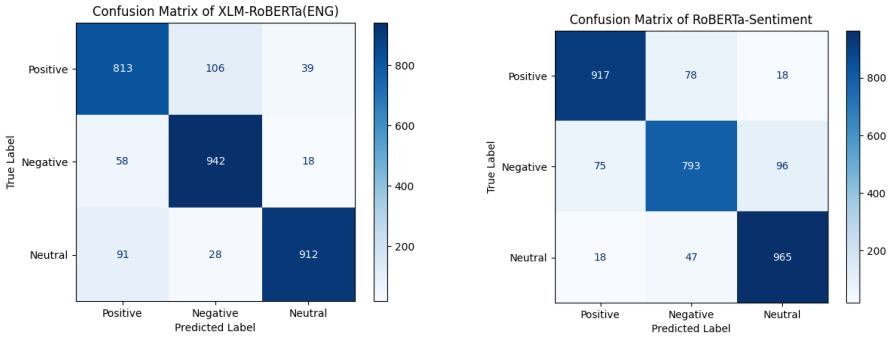


Figure 4.4: Confusion matrix of Transformer-based in English dataset

From Fig 4.3 we can see, BanglaBERT has the highest correct prediction. After that XLM-RoBERTa has the second highest correct prediction rate. On the other hand, RoBERTa-Sentiment has the highest correct prediction rate for the English Dataset.

4.2.3 Performance Metrics:

In Fig 4.5 and 4.6, we can see the best performed model performance metrics. BanglaBERT is the best model for our Bengali dataset and RoBERTa-Sentiment outperformed. So these two models are best for our dataset.

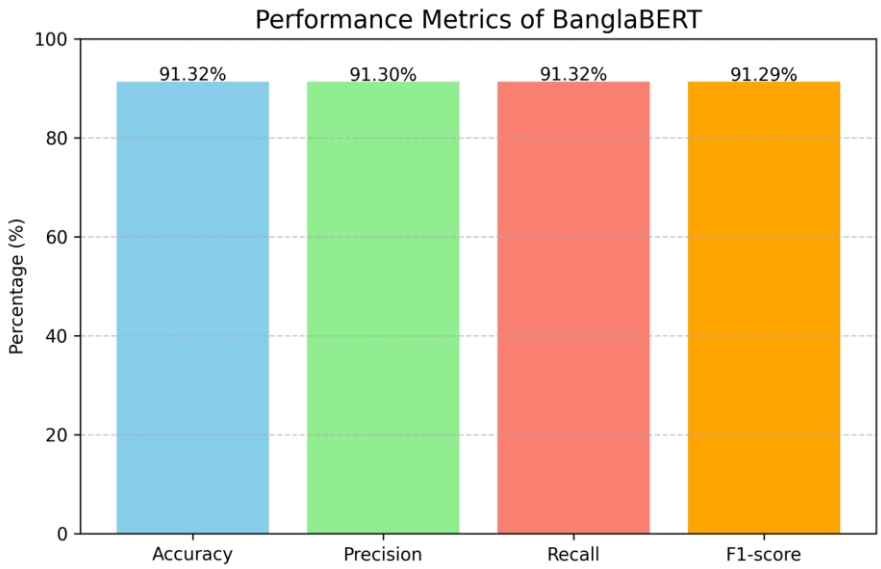


Figure 4.5: Performance metric of BanglaBERT

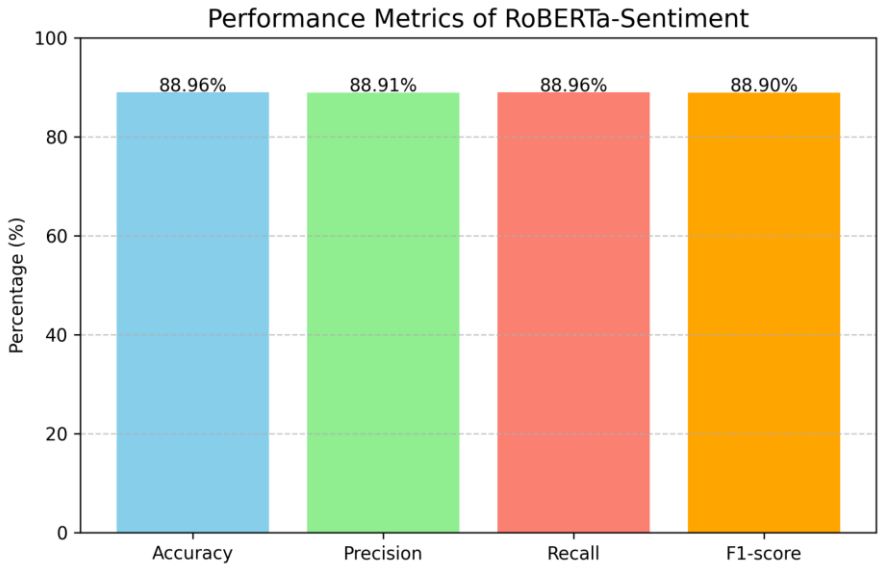


Figure 4.6: Performance metric of RoBERTa-Sentiment.

4.2.3 Explanation of XAI-based LIME:

LIME accomplishes this by interpreting a prediction of a black-box model through local approximation. When making a prediction, LIME creates numerous perturbed samples by making small changes to the initial input and recording the corresponding output of the black-box model. It then applies these perturbed examples and predictions to train an interpretable and solvable surrogate model, such as linear regression or a decision tree, that simulates the behavior of the locally complex model around the example. Finally, the surrogate model determines the most important features that contributed to the prediction, providing a human-interpretable justification for how the choice was made.

$$argmingL(f, g, \pi x) + \Omega(g)$$

From the 4.7, we can see that our XAI-based LIME gives this explanation for "পণ্যটি ভালো কিন্তু ভাঙা ছিল" review.

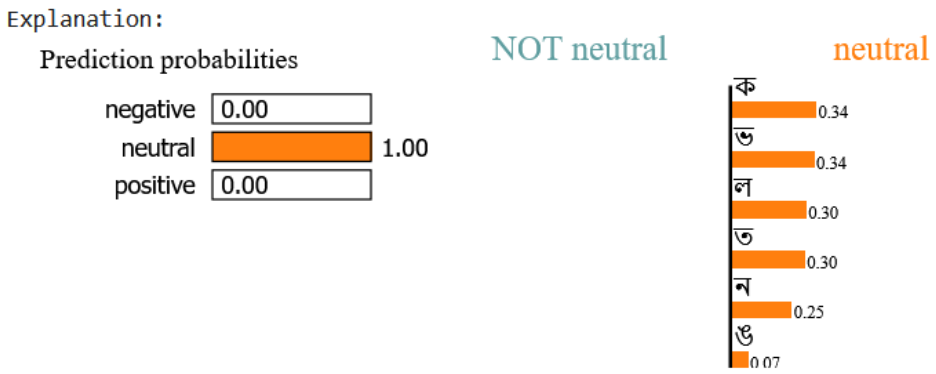


Fig 4.7: Example XAI-based LIME output for BanglaBERT.

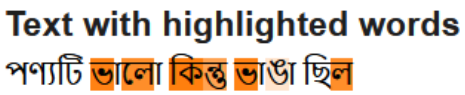
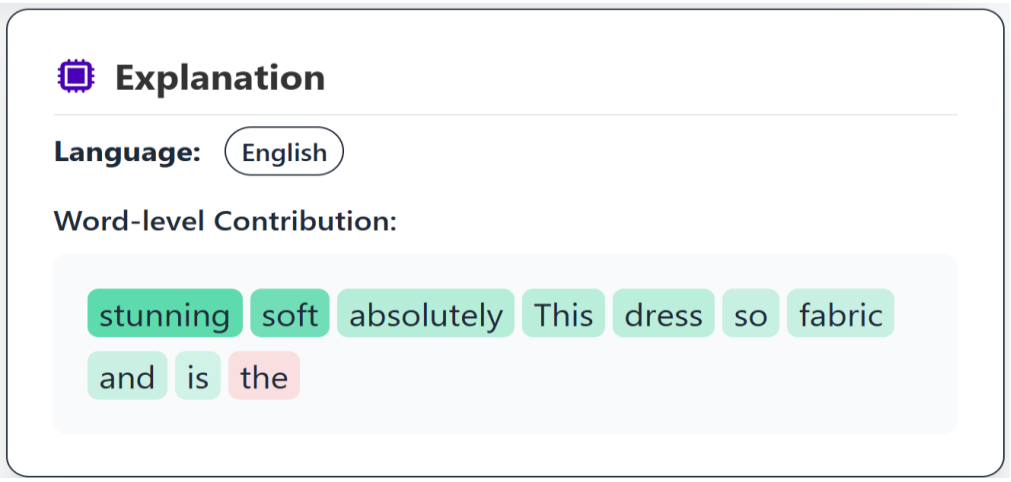


Figure 4.8: Priority of word sequences in each paragraph using LIME for BanglaBERT.

In the review, each word comes in a sequence. If not in order, then the sentences in the paragraph will be wrong, and the data will not be trained properly. In Figure 11 "পণ্যটি ভালো কিন্তু ভাঙা ছিল" which is "The product was good but broken.", in which we can see the priority-based words and characters that are predicted nationally, and the orange region predicts the character weight.

Another Example of LIME(XAI) for the review "**This dress is absolutely stunning and the fabric is so soft**"



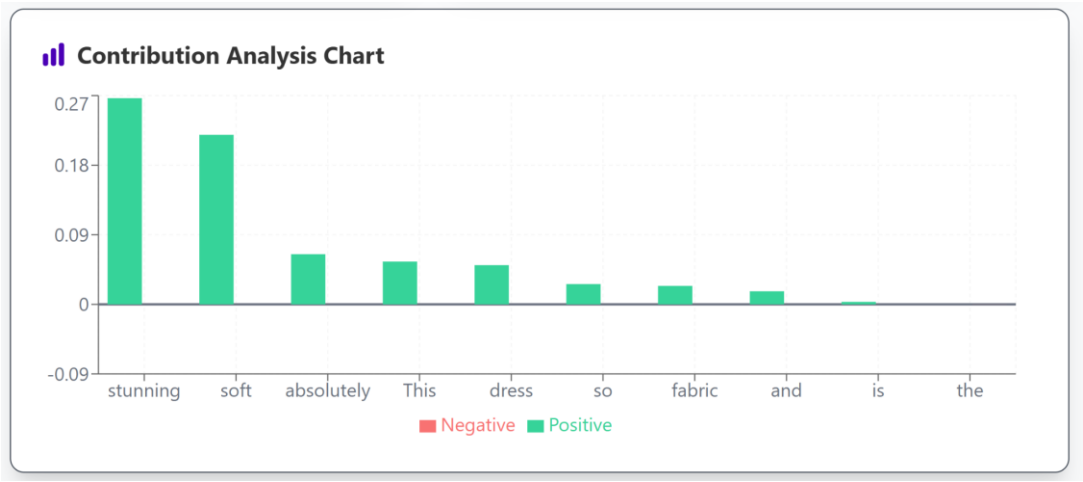


Figure 4.9: Priority of word contribution and weight score chart using LIME

4.1 Summary

This chapter describes the complete implementation and evaluation process of the multilingual sentiment analysis system. It started with the environment setup, where the tools, libraries, and hardware used for data preprocessing and model training were specified. The metrics used to evaluate model performance during the testing and evaluation stages - precision, accuracy, recall, and F1-score - were presented. The comparative analysis showed that the Transformer models significantly outperformed the deep learning models, with BanglaBERT achieving the best results for Bangla data and RoBERTa-Sentiment achieving the best results for English data. The results and discussion highlight the power of Transformers in handling linguistic complexity and multilingual context, while LIME improved the interpretability by explaining the prediction results. Overall, the chapter demonstrated that Transformer-based models provide the most effective solution for multilingual sentiment classification in e-commerce reviews.

Chapter 5

Engineering Standards and Design Challenges

Standards in engineering validate reliability, safety, and interoperability in the design of a
system. Concurrently, a design challenge emerges from solving intricate real-world issues and complying with standards in an effective way.

5.1 Compliance with the Standards

The proposed system is web-based and software-intensive and relies on transformer models and deep learning. Adherence to communication standards and software protocols is, therefore, more relevant than hardware-specific protocols. For each standard, there is an analysis of alternatives in terms of advantages and disadvantages, with explanation of the chosen option.

5.1.1 Software Standards

Throughout the realization of this project, software standards were utilized to assist in ensuring consistency, maintainability, and overall quality of the system. Of the many available standards, IEEE 830 – Software Requirements Specification (SRS) and ISO/IEC 9126 – Software Product Quality were employed as guidelines. IEEE 830 provided a formal template for documenting requirements in an unambiguous, clear way and for making sure that both functional and non-functional aspects of the system were properly defined. This was particularly useful for a research-focused project because it enabled a balance to be maintained between academic and implementation pragmatism. On the other hand, ISO/IEC 9126 was employed as a benchmark against which the quality of the system under development could be evaluated. By taking heed of such qualities as functionality, reliability, usability, efficiency, and maintainability, the project adhered to international software quality standards.
Other methods were also considered. For instance, Agile-based informal documentation could have provided more flexibility and faster iterations, especially in the prototyping phase. However, the lack of formal structure in Agile documentation has the tendency to develop inconsistencies, which may be unsuitable for academic research where reproducibility and clarity are most essential. The second alternative was to utilize purely custom-made academic documentation, developed specifically for the thesis. While it would have offered maximum flexibility, it lacked global recognition and comparability with industry software practice.

By adhering to IEEE and ISO standards, the project thus not only guaranteed proper communication of its requirements but also guaranteed that its quality could be assessed on the basis of globally accepted parameters. This choice thus set the system on a professional pedestal, making it academically sound and practically viable.

5.1.2 Hardware Standards

Though the system suggested here is mostly software-based, hardware standards are necessary to ensure train and inference reproducibility and model reliability. The most relevant standard in this case is IEEE 754 – Floating Point Arithmetic, which defines the representation and behavior of floating-point numbers in computer systems. Deep learning frameworks such as TensorFlow and PyTorch utilize IEEE 754 precision heavily to carry out large-scale matrix multiplications and gradient calculations. By adhering to this standard, the system ensures that computation is maintained consistently across a wide range of hardware environments, whether executed on CPUs, GPUs, or cloud-based accelerators. This is particularly important for sentiment analysis, where minute numerical differences will affect model reproducibility.

Alternative approaches were also explored. For example, low-precision computing methods such as INT8 or FP16 quantization are increasingly being used to speed up inference and reduce computational costs, especially in resource-constrained environments. While these methods are highly efficient and power-saving, they can sometimes pose a compromise on model accuracy, especially in multilingual sentiment classification where the fine-grained linguistic distinctions have to be preserved. The alternative was to rely entirely on custom hardware optimizations such as Tensor Cores in NVIDIA GPUs, which are quicker but at the cost of portability across different hardware.

After consideration, the project adhered to IEEE 754 as it offers the best tradeoff between performance, compatibility, and accuracy. While quantization and hardware-specific optimizations could be considered for future deployment, IEEE compliance ensured that the outcome of the research was consistent, portable, and easily reproducible on a range of computational platforms, which is a necessity in an academic and experimental setting.

5.1.3 Communication Standards

Communication standards were also a requirement in this project to offer effective interaction among the frontend app and backend services. The system used RESTful API conventions using HTTP/1.1 with JSON encoding as the primary communication standard. This was chosen because REST APIs are lightweight, highly supported, and easy to incorporate into modern web apps. Utilizing JSON as the data exchange format allowed it to be used with both the React.js frontend and FastAPI backend and made the system highly human-readable and developer-friendly.

Other options were weighed, including GraphQL and gRPC. GraphQL is more adaptive since clients can declare specifically the information they need in a query to reduce unnecessary network requests. However, it adds to complexity in schema definition and can end up being over-engineered for a relatively simple input-output system like sentiment classification. Similarly, gRPC is high performance and efficient with binary communication but has the requirement for special client libraries and does not support easy support in web browsers without additional layers.

In light of these factors, RESTful APIs with JSON were selected as the most suitable communication standard. They were the right mix of simplicity, interoperability, and scalability, so that the system could be accessed by the user without excessive technical baggage.

5.2 Impact on Society, Environment and Sustainability

The proposed sentiment analysis system has profound impacts on life, society, and the environment. On the individual level, the system enhances user experience by simplifying product review analysis for making the right purchasing decision. For enterprise, the system provides valuable customer insights for better service delivery and competitiveness in the market. On the societal level, the system promotes digital inclusion via Bangla and Banglish text support, boosting local e-commerce participation. Environmentally, the large-scale deep learning models are challenging in terms of high energy consumption during training. In an effort to address sustainability issues, the project focuses on reusing pre-trained models, having efficient hardware practices, and creating light alternative versions for future deployment.

5.2.1 Impact on Life

The system directly impacts everyday life by making e-shopping more reliable and convenient. For consumers, it reduces the time and effort spent on reading long reviews by automatically labeling them as positive, negative, or neutral in an instant. It helps individuals make quicker and more confident purchasing decisions. For businesses, the program provides them with a sense of customer opinion, enabling them to improve product quality and customer service. Finally, the system renders online marketplaces more reliable and convenient.

5.2.2 Impact on Society & Environment

The system makes society better by facilitating digital literacy in Bangla and empowering local e-commerce ecosystems. The system helps companies in responding to customer demands more effectively, instilling trust in online systems. Environmental concerns can be invoked due to the enormous computational cost of training deep learning models, which consume much energy and contribute to carbon emissions.

5.2.3 Ethical Aspects

The system's design has ethical considerations at its foundation. User privacy is upheld in the form of non-storage of sensitive or personal data, consistent with GDPR-like principles. Sentiment analysis bias remains a problem that still persists, where models tend towards one language or dialect over another. The system was therefore tested on Bangla, English, and Banglish datasets to test for fairness. Transparency of methodology and appropriate use of AI enable the system to be ethically implemented and minimize the potential for abuse.

5.2.4 Sustainability Plan

The system is made sustainable by making informed design choices that balance performance and resource efficiency. Instead of training large models from scratch, the project taps into already trained models like BanglaBERT and XLM-RoBERTa, which is time-saving as well as energy-efficient. Future work would entail application of model distillation and quantization techniques such that smaller versions of the models can be run on limited hardware at minimal loss of accuracy. Deployment into the cloud will be centered around energy-efficient servers with modular architecture so that newer models without pollution can be added at minimal inconvenience. All these steps lead to long-term sustainability and reduce environmental footprints.

5.3 Project Management and Financial Analysis

The project was controlled based on a structured management process, with the dataset preparation, preprocessing, model development, evaluation, and deployment phases. Task allocation tables and Gantt charts were employed to track progress, and to make sure that responsibilities and timelines were understood.

5.3.1 Cost Analysis:

Table 5.1: Estimated cost table.

Category	Item/Resource	Estimated Cost (Taka)
Hardware	Laptop with GPU	60,000-70,000 Tk
Software & Tools	Python, Colab, Libraries	0 Tk
Data Collection & Preprocessing	Internet cost, Web scraping	5000-6000 Tk
Miscellaneous	Printing, Documentation, Report Binding	500-1000 Tk
Total Estimated Cost		65,500 - 77,000 Tk

5.3.2 Revenue Model

If pursued beyond academic scope, the system can have revenue streams via:
A subscription model for businesses, offering premium sentiment analysis and analytics dashboards. A freemium model where individual users enjoy free basic sentiment classification, and businesses pay for premium offerings such as bulk analysis and API integration. This model ensures both accessibility to individuals and viability for business use at scale.

Table 5.2: Details of Tools and Platforms Used

Category	**Tool/Platform**	**Purpose**
Programming Language	Python	Core NLP and DL, Transformers Development
Libraries/Frameworks	Hugging Face, Transformers	Model training, RAG integration, embeddings
Model Training	Google Colab (GPU)	Scalable training and testing
IDE	Google Colab	Experiment tracking and visualization
Version Control	GitHub	Code versioning and collaboration
Storage/Backup	Google Drive	Dataset storage and sharing
Web Framework	Vite + React.js	Website Development

5.4 Complex Engineering Problem

5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.1). For P2, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.3: Mapping with Complex Engineering Problem.

EP1 Dept of Knowledge	EP2 Range Of Conflicting Requirements	EP3 Depth of Analysis	EP4 Familiarity of Issues	EP5 Extent of Applicable Codes	EP6 Extent Of Stake-holder Involvement	EP7 Interdependence
✓	✓	✓	✓	✓	✓	✓

Rationale for Mapping

EP1 (Depth of Knowledge): The project demands specialist knowledge in machine learning, transformer models, and multilingual NLP, going beyond basic programming.

EP2 (Conflicting Requirements): The system must balance performance with cost and sustainability—higher accuracy often requires more expensive resources.

EP3 (Depth of Analysis): Analysis includes TF-IDF, deep learning, transformer fine-tuning, and evaluation with multiple metrics.

EP4 (Familiarity of Issues): While English sentiment analysis is common, Bangla and Banglish analysis is an emerging field with limited research.

EP5 (Applicable Codes): No single universal code exists; IEEE 830, ISO/IEC 9126, and ethical AI practices are applied.

EP6 (Stakeholder Involvement): Stakeholders include consumers, businesses, researchers, and future developers extending the system.

EP7 (Interdependence): Success depends on coordination across data preprocessing, model inference, backend APIs, and frontend UI design.

Mapping with Knowledge Profile for EP1

This table 5.2 is designed to map the EP1 to the Knowledge Profile.

Table 5.4: Mapping with knowledge Profile.

K3 Engineering Fundamentals	K4 Specialist Knowledge	K5 Engineering Design	K6 Engineering Practice	K8 Research Literature
✓	✓	✓	✓	✓

Rationale for Knowledge Profile Mapping

K3: The project relies on core ML fundamentals such as embeddings, optimization, and evaluation metrics.

K4: Specialist NLP techniques (Bangla stemming, tokenization, transformers) are required.

K5: The system is an engineered design with modular architecture (preprocessing → models → API → UI).

K6: Practical skills in coding, deployment, and integration make this a real-world system.

K8: Research papers and pretrained models formed the basis for developing a robust solution.

5.4.2 Engineering Activities

The development process of the multilingual sentiment analysis system entailed a wide range of engineering activities such as research, design, implementation, and evaluation. The activities were crucial to achieve both the practical and academic objectives of the project.

Table 5.5: Mapping with Complex Engineering Activities.

EA1 Range of re- sources	EA2 Level of Interaction	EA3 Innovation	EA4 Consequences for society and environment	EA5 Familiarity
✓	✓		✓	✓

5.5 Summary

This chapter has examined the engineering standards, design problematics, and broader implications of the multilingual sentiment analysis system. It underscored conformance to relevant software, hardware, and communication standards and justified why they were selected out of available options with well-stated advantages and disadvantages. The chapter continued with the implications of life, society, ethics, and sustainability and demonstrated how the system enhances users' experiences in addition to resolving environmental and fairness concerns. A sustainability plan was charted to ensure long-term efficiency via modularity and more sustainable model choices. Lastly, the chapter addressed project management and finances with alternative budget proposals, why each would be chosen, and a feasible revenue model. In so doing, they determined the project with professional engineering practice along with real-world issues on cost, ethics, and sustainability.

Chapter 6

Conclusion

In conclusion, we are gonna summarize the whole study, we are gonna talk about limitations of our work and the future plans of the study.

6.1 Summary

In this work, we investigated sentiment analysis of Bengali and English e-commerce product reviews using merged deep learning (CNN, LSTM, BiLSTM, CNN–LSTM) and transformer (XLM-RoBERTa, BanglaBERT, RoBERTa- Sentiment) models for multilingual sentiment classification. Experimental results reflected that transformer models were superior to deep learning models in all cases, where BanglaBERT was superior for Bengali reviews and RoBERTa-Sentiment was superior for English reviews. The utilization of LIME created interpretability and brought in transparency and trust in model outputs. Generally, the research helps improve NLP research on low-resource languages and provides useful knowledge for e-commerce applications.

6.2 Limitation

While the study found encouraging results, there are some limitations. The corpus contained only a portion of reviews (≈15,000 for Bengali) and needed to be manually preprocessed with Banglish and code-mixed text, so no scaling was feasible. Computational resources limited large-scale experimentation and heavy hyperparameter tuning. Additionally, class skewing and domain dependence of reviews could have affected performance.

6.3 Future Work

Future research can improve this study in the following ways. First, the dataset can be expanded and balanced, and code-mixed and Bengali text processing algorithms can be developed. Second, model compression and optimization methods can be used to reduce computational costs and facilitate real-time deployment. Third, we will try to apply more models hopefully. We have a plan to implement a Large Language Model(LLM) in the future.

References

[1] R. K. Das, M. Islam, M. M. Hasan, S. Razia, M. Hassan, and S. A. Khushbu, "Sentiment analysis in multilingual context: Comparative analysis of machine learning and hybrid deep learning models," Heliyon, vol. 9, no. 9, 2023.

[2] Z. Ahmed, S. S. Shanto, and A. I. Jony, "Advancement in Bangla sentiment analysis: A comparative study of transformer-based and transfer learning models for e-commerce sentiment classification," *Journal of Information Systems Engineering & Business Intelligence*, vol. 9, no. 2, 2023.

[3] S. Kusal, S. Patil, A. Gupta, H. Saple, D. Jaiswal, V. Deshpande, and K. Kotecha, "Sentiment analysis of product reviews using deep learning and transformer models: A comparative study," in Proc. Int. Conf. Artif. Intell. Textile and Apparel, Singapore: Springer Nature Singapore, Aug. 2023, pp. 183–204.

[4] Kabir, A. Masud, M. Hasan, S. Sunny, M. A. Pramanik, and S. Chakraborty, "A transformer-based approach for analyzing scrapped e-commerce product reviews from social media platforms with explainable AI," in Proc. 2024 27th Int. Conf. Computer and Information Technology (ICCIT), Dec. 2024, pp. 2110–2115.

[5] S. S. Shanto, Z. Ahmed, N. Hossain, A. Roy, and A. I. Jony, "Binary vs. multiclass sentiment classification for Bangla e-commerce product reviews: A comparative analysis of machine learning models," International Journal of Information Engineering and Electronic Business (IJIEEB), vol. 15, no. 6, pp. 48–63, 2023.

[6] R. Chowdhury, F. U. Zaman, A. Sharker, M. Rahman, and F. M. Shah, "Rate insight: A comparative study on different machine learning and deep learning approaches for product review rating prediction in Bengali language," in Proc. 2022 25th Int. Conf. Computer and Information Technology (ICCIT), Dec. 2022, pp. 406–411.

[7] O. Bellar, A. Baina, and M. Ballafkih, "Sentiment analysis: Predicting product reviews for E-commerce recommendations using deep learning and transformers," Mathematics, vol. 12, no. 15, p. 2403, 2024.

[8] T. Alsaedi, M. R. R. Rana, A. Nawaz, A. Raza, and A. Alahmadi, "Sentiment mining in E-commerce: The transformer-based deep learning model," International Journal of Electrical and Computer Engineering Systems, vol. 15, no. 8, pp. 641–650, 2024.

[9] N. R. Bhowmik, M. Arifuzzaman, and M. R. H. Mondal, "Sentiment analysis on Bangla text using extended lexicon dictionary and deep learning algorithms," Array, vol. 13, p. 100123, 2022.

[10] M. S. Islam and K. M. Alam, "Sentiment analysis on Bangla food reviews using machine learning and explainable NLP," in *Proc. 2023 26th Int. Conf.*

Computer and Information Technology (ICCIT), Dec. 2023, pp. 1–6.

[11] S. N. Nobel, S. M. R. Swapno, H. H. Shajeeb, M. B. Islam, and R. Haque, "Hybrid CNN LSTM approach for sentiment analysis of Bengali language comment on Facebook," in Proc. 2024 Int. Conf. Integrated Circuits and Communication Systems (ICICACS), Feb. 2024, pp. 1–8.

[12] P. Savci and B. Das, "Prediction of the customers' interests using sentiment analysis in e-commerce data for comparison of Arabic, English, and Turkish languages," *Journal of King Saud University - Computer and Information Sciences*, vol. 35, no. 3, pp. 227–237, 2023.

[13] M. A. Rahman, M. Begum, T. Mahmud, M. S. Hossain, and K. Andersson, "Analyzing sentiments in e-learning: A comparative study of Bangla and Romanized Bangla text using transformers," IEEE Access, vol. 12, pp. 89144–89162, 2024.

[14] M. N. Hoque, U. Salma, M. J. Uddin, M. M. Ahamad, and S. Aktar, "Exploring transformer models in the sentiment analysis task for the under-resource Bengali language," Natural Language Processing Journal, vol. 8, p. 100091, 2024.

[15] M. Hoq, P. Haque, and M. N. Uddin, "Sentiment analysis of Bangla language using deep learning approaches," in Proc. Int. Conf. Computing Science, Communication and Security, Cham: Springer International Publishing, Feb. 2021, pp. 140–151.

[16] S. Sunny, S. Pinky, S. Jalal, M. Kayser, M. Wadud, and N. Mansoor, "Bangla e-commerce sentiment analysis optimization using tokenization and TF-IDF," in Proc. 2024 Int. Conf. Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Mar. 2024, pp. 1–6.

[17] M. R. A. Rashid, K. F. Hasan, R. Hasan, A. Das, M. Sultana, and M. Hasan, "A comprehensive dataset for sentiment and emotion classification from Bangladesh e-commerce reviews," Data in Brief, vol. 53, p. 110052, 2024.

[18] M. Ahmed Masum, S. Junayed Ahmed, A. Tasnim, and M. Saiful Islam, "BAN-ABSA: An aspect-based sentiment analysis dataset for Bengali and its baseline evaluation," in Proc. Int. Joint Conf. Advances in Computational Intelligence: IJCACI 2020, Singapore: Springer Singapore, May 2021, pp. 385–395.

[19] M. M. Alam, A. Shome, S. Saha, and M. F. Mridha, "BRevML: Classifying Bangla reviews for e-commerce using machine learning," in Proc. 2021 Int. Conf. Science & Contemporary Technologies (ICSCT), Aug. 2021, pp. 1–6.

[20] S. Sazzed, "BengSentiLex and BengSwearLex: Creating lexicons for sentiment analysis and profanity detection in low-resource Bengali language," PeerJ Computer Science, vol. 7, p. e681, 2021.

[21] S. Ahmed, M. M. Samia, M. H. Sayma, M. M. Kabir, and M. F. Mridha, "tRF-BERT: A transformative approach to aspect-based sentiment analysis in the Bengali language," PLoS One, vol. 19, no. 9, p. e0308050, 2024.

[22] K. Taneja, J. Vashishtha, and S. Ratnoo, "Transformer based unsupervised learning approach for imbalanced text sentiment analysis of e-commerce reviews," Procedia Computer Science, vol. 235, pp. 2318–2331, 2024.

[23] S. Shan, J. Sun, and R. M. C. Macawile, "Examining customer satisfaction through transformer-based sentiment analysis for improving bilingual e-commerce experiences," IEEE Access, 2025.

[24] H. Huang, A. A. Zavareh, and M. B. Mustafa, "Sentiment analysis in e-commerce platforms: A review of current techniques and future directions," *IEEE Access*, vol. 11, pp. 90367–90382, 2023.

[25] T. Mahmud, R. Karim, R. Chakma, T. Chowdhury, M. S. Hossain, and K. Andersson, "A benchmark dataset for cricket sentiment analysis in Bangla social media text," Procedia Computer Science, vol. 238, pp. 377–384, 2024.

[26] K. O'Shea and R. Nash, "An introduction to convolutional neural networks," arXiv preprint arXiv:1511.08458, 2015.

[27] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[28] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," IEEE Transactions on Signal Processing, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[29] Y. Donahue, L. A. Hendricks, S. Guadarrama, M. Rohrbach, S. Venugopalan, K. Saenko, and T. Darrell, "Long-term recurrent convolutional networks for visual recognition and description," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2015, pp. 2625–2634.

[30] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, "Unsupervised cross-lingual representation learning at scale," in *Proc. 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, Jul. 2020, pp. 8440–8451.

[31] M. F. Imam, A.-A. Nahid, and M. S. Islam, "BanglaBERT: A pre-trained language model for Bangla language understanding," *arXiv preprint arXiv:2106.05974*, 2021.

[32] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," arXiv preprint arXiv:1907.11692, 2019.

[33] 3.1: GeeksforGeeks, “NonFunctional requirements in software engineering,”

GeeksforGeeks, Jul. 15, 2025. https://www.geeksforgeeks.org/software-engineering/non-functional-requirements-in-software-engineering/

[34] 3.2: GeeksforGeeks, "Context diagrams," GeeksforGeeks, Mar. 21, 2024. https:// www.geeksforgeeks.org/system-design/context-diagrams/#what-are-context-diagrams